Mind Map: Immune Response to Extracellular Bacterial Infection

Summary Points

- Most immune cells are of hematopoietic origin
- Innate immunity: granulocytes, monocytes/macrophages, dendritic cells, NK cells
- Adaptive immunity: T and B lymphocytes
- Antigen-presenting cells: dendritic cells, monocytes/macrophages, B lymphocytes
- Other immunocompetent cells: epithelial cells, endothelial cells, platelets
- Primary lymphoid organs: bone marrow, thymus
- Secondary lymphoid organs: spleen, lymph nodes, mucosa-associated lymphoid tissue (MALT)
- Secondary lymphoid organs are sites of immune responses

Step 1: Barrier Breach

- Epidermis as physical barrier
- Competition with commensal flora
- Antibacterial peptides and enzymes
- Barrier breach (cut, puncture)

Step 2: Pathogen Recognition

- Innate immune cells in subcutaneous tissue (macrophages, immature dendritic cells)
- Recognition of pathogen-associated molecular patterns (PAMPs)
- Danger signals

Step 3: Phagocytosis

- Internalization of bacteria by macrophages and dendritic cells
- Initiation of inflammatory response

Step 4: Vascular Changes

- Increased vascular permeability
- Movement of immune cells and proteins through endothelium
- Neutrophils as key players in bacterial elimination

Step 5: Dendritic Cell Maturation

- Immature dendritic cells receive danger signals
- Maturation and migration to secondary lymphoid organs

Step 6: Antigen Presentation

- Interaction with adaptive immune cells (B and T cells)
- Activation of B and T cells

Step 7: Clonal Expansion

- Proliferation of activated B and T cells
- Formation of memory lymphocytes

Step 8: Plasma Cell and Antibody Production

- Differentiation of B cells into plasma cells
- Production of antibodies against bacterial proteins

Step 9: Antibody Function

- Antibodies travel through the bloodstream
- Direct bacterial destruction via complement activation
- Enhanced phagocytosis by macrophages

Step 10: Tissue Repair

- Elimination of bacteria
- Processes involved in restoring epithelial integrity
- Restoration of barrier protection

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